

Rachel Ramsey IDC 6940

Week2 - Literature Review

• Background/Motivation:

– Describe the context in which the research was conducted.

– OLS is the current standard in macroeconomic forecasting.

– SGDR regressor has been a part of recent attempts to make reports more current and account for changing circumstances as well as computational costs.

– What problem or gap in the existing literature does the paper aim to address?

The author compares linear, ensemble and deep-learning approaches on the same dataset. This addresses the difference in accuracy and outcomes when using different methods.

– Discuss the significance of the research question and why it is important.

– Predicting unemployment data can give the public a more current picture of the economy because current reports lag by at least a month. It also allows for a simulation option by changing input variables to match policy options.

• Methods Used:

– Summarize the methodologies employed by the authors.

The authors build a machine learning process where they take BLS historical unemployment data, preprocess it and

run it through a variety of algorithms to determine the most accurate forecast. The authors use linear, non-linear, and deep learning algorithms to cover more complexity and provide a wide range of results for a good comparison.

– Explain how these methods are suited to address the research question.

The research question for this paper focuses on which algorithm produces the most accurate prediction.

These methods address this focus by using a variety of different types of algorithms in order to cover any complexities in the data like non-linear relationships or high-dimensionality.

– Discuss any innovative approaches or techniques that are particularly noteworthy.

The method of this paper being machine learning techniques in economic data is innovative because the field of economics has a subfield of econometrics which is used to forecast. Econometrics has a focus of understanding what factors contribute to an outcome while machine learning has the goal of predicting without as

much concern for how or what factors contributed to the outcome.

I find this approach of using machine learning to focus on accuracy of the prediction rather than causal effects to be noteworthy.

• Significance of the Work:

– Highlight the key findings and contributions of the paper.

The paper compared seven algorithms and six different scalars. In terms of the algorithms, tree based ensembles outperformed all other algorithms. The author states that the Random Forest algorithm had the lowest Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE) and that this can be traced to its ability to

handle complexities such as nonlinear interactions.

MaxAbsScaler was determined to be the best scalar because it provided the best preprocessing for all models among tested Scalars.

– Explain why the results are important within the broader context of the field.

These results are important for the field of economics because they give guidance as to where the field could go in the future. The practice of forecasting in economics has been a lagged process and the ability to release real-time forecasts could mean a lot for the future of economic policies and personal

economic choices.

– Discuss the implications of the findings for future research or practice.

The future of economic policy and personal economic choices can be changed by real-time forecasting. The 2008 financial crisis in the US could have been less detrimental to many individuals and families across the country if there were more accurate information early enough to make better decisions or avoid making mistakes. The same could be said for economic policies. The Fed could be making more informed choices as well as presidential cabinets when it comes to the needs of the country's economy.

• Connection to Other Work:

– Relate the paper to other relevant studies.

– How does this paper build on or differ from previous work?

The author mentions several papers where work had been done using the features used in this paper. The author uses these examples as support for their use of them. The author also mentions an example where each model was used in economics. This paper builds on these examples by comparing all the models.

– Identify any references to seminal works or influential papers cited by the authors.

The author refers to several paper such as:

- Stock & Watson (2001): OLS in economic analysis
- Breiman (2001): Random Forest
- Drucker et al. (1997): SVR
- Hochreiter & Schmidhuber (1997): LSTM Network

The following were used to highlight the features used in this paper:

- Liu & Zhang (2018)
- Chauvet & Potter (2013)

• Relevance to Capstone Project:

– Discuss how the content of the paper might be relevant to your own capstone project.

This paper uses some of the same data sources I plan to use such as BLS data.

The author is also using machine learning in the economics field which is the goal of my paper.

– Identify any specific methods, theories, or findings that you might incorporate into your project.

I believe it may be beneficial to use the findings from this paper in my project. Using the model and scaler that were most successful in this paper may lead me down a successful path in terms of the goal of my project.

– Highlight any potential areas where your capstone could expand upon or diverge from the paper's findings.

I believe my paper may need to diverge from the findings of this paper because the model that was most successful was deemed successful when using unemployment data and I plan to use occupational data.